

Mixed-Length Shapelets to KMeans

Pipeline and Explanation Stage

Explainable Clustering for Spatio-Temporal Data

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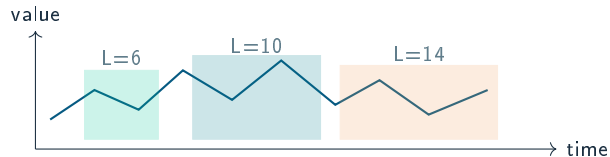
What This Deck Explains

- How mixed-length shapelets are sampled and selected.
- How distances to shapelets become a fixed-width feature vector.
- How KMeans is selected and fit on the shapelet-distance matrix.
- How explanations are derived from surrogate + SHAP + abductive edits.

Code anchors

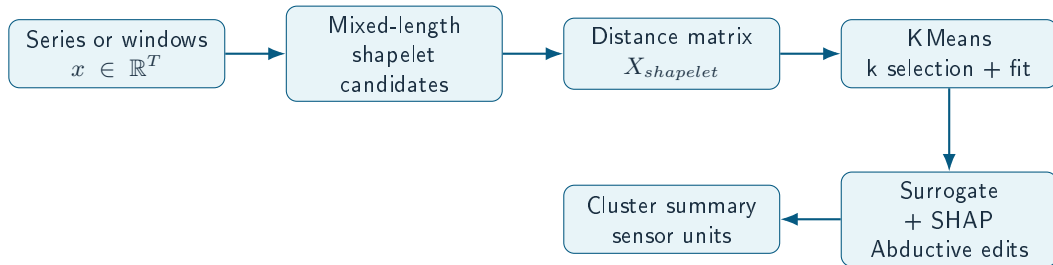
```
scripts/research/  
shapelet_stability.py  
scripts/notebooks/  
shapelets_ecg200.ipynb  
shapelets_ecg5000.ipynb  
shapelets_roma_taxi.ipynb
```

Mixed-Length Shapelets: Visual Intuition



- Candidate shapelets are sampled from multiple lengths.
- Mixed lengths capture short and long motifs in one dictionary.
- Each shapelet becomes one feature after distance transform.

Pipeline Overview (Data Flow)



Mixed-length sampling gives one dictionary; distances make it fixed-width.

KMeans runs on that matrix; explanations follow.

Step 1: Build a Mixed-Length Dictionary

- Input: $x \in \mathbb{R}^{N \times T}$, length list (e.g., (5, 10, 15, 20)).
- For each candidate, randomly choose length, series, and start index.
- Code: `extract_random_shapelets` in `shapelet_stability.py`.

```
for cand in
1..n_candidates:
    L = choice(lengths)
    row = random series
    start = random index
    s = x[row, start:start+L]
```

Mixed lengths are explicit in the `lengths` list and sampled per candidate.

Step 2: Shapelet-Distance Feature Matrix

- Each shapelet yields one feature: distance from a series to that shapelet.
- Implementation: `min_distance_to_shapelet + compute_shapelet_distances`.
- Standardize the distance matrix before KMeans.

Distance mapping

$$\phi_j(x) = \min_t \|x[t : t + L_j] - s_j\|_2$$

$$X_{shapelet} \in \mathbb{R}^{N \times m}$$

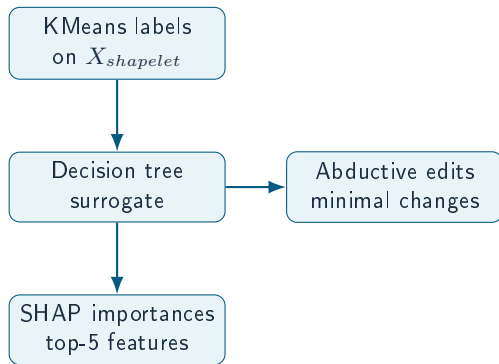
Mixed-length shapelets still produce a fixed-width matrix because each shapelet contributes one column.

Step 3: KMeans Selection and Clustering

- Evaluate k on shapelet features with multiple seeds.
- Metrics: silhouette and stability ARI across seeds.
- Composite score: `silhouette + 0.5 * stability_ari`.
- Code: `search_best_k` + final KMeans fit.

```
for k in k_grid:
    labels =
    KMeans(k).fit_predict(X)
    sil = silhouette(labels)
    stability = mean(ARI)
    score = sil + 0.5*stability
```

Step 4: Explanation Stage (Transparent Outputs)



- Surrogate: `DecisionTreeClassifier` on shapelet features.
- Fidelity: cross-validated accuracy vs KMeans labels.
- Stability: top-5 feature overlap across seeds.
- Abductive: minimal feature edits that flip cluster.

Supervisor-Ready Outputs

- Slide deck: `docs/slides/2026-05-27/shapelets_mixed_length_kmeans.pdf`.
- Pipeline notes: `docs/SHAPELETS_WORK_NOTES.md`.
- Core implementation: `scripts/research/shapelet_stability.py`.
- Dataset-specific notebooks for figures and runs: `shapelets_ecg200`, `shapelets_ecg5000`, `shapelets_roma_taxi`.

Use the pipeline diagram and Step 2/3 slides to explain the mixed-length to KMeans transition clearly.